CISC 699 — Assignment 1

Project Launch Packet (Thesis / Research Track)

Working Title:

AI-Driven Cybersecurity Risk Management: A Systematic Review of Emerging Threats, Security Controls, and Governance Challenges

Student: [Your Name]

Program Track: Thesis (Research) — not a technical build/artifact track

Date: [Submission Date]

Quick note before you read this: I've updated it a few times already, once after the professor's class-wide feedback on the first ungraded round, and again after I wrote Discussion 1 (Ethics Readiness and Evidence of Preparation), which turned up governance frameworks and ethics issues that really belonged here too. The full history of what changed and why is in the AI Usage Log at the end.

1. Problem Statement

Companies are rolling out AI faster than their security teams can really keep up with it: machine-learning intrusion detection, generative AI copilots, automated fraud detection, that kind of thing. The people stuck in the middle of this are usually CISOs, IT risk managers, and compliance officers. They have to say yes to these projects because of the upside (better detection, more efficiency), but they're also the ones who get blamed when the AI system opens up some new attack surface, leaks data it shouldn't, or just does something nobody expected in production. This isn't a hypothetical scenario I made up for a thesis proposal. It's happening in real organizations right now.

It's not that nobody's researching this. There's actually a lot of work out there on what can go wrong with AI, adversarial attacks, data poisoning, deepfakes used for social engineering, all of it. What's missing is the next step. Once you know about a risk, how do you actually pick the right controls for it, and then get that decision folded into governance and compliance in a way a board or regulator would accept? Nobody's really answered that yet, and part of the reason is just timing. The big governance references, the NIST AI Risk Management Framework, ISO/IEC 42001, NIST's Cybersecurity Framework 2.0, the EU AI Act, are all only two or three years old, so there hasn't been time to line them up against the threat-and-controls research. That gap, between knowing about a threat and knowing how to actually govern it, is basically why this thesis exists.

2. Candidate Research Directions

I didn't just run with the first idea I had. I looked at three different directions before picking one, all somewhere in the AI-and-cybersecurity space, but pretty different from each other in scope and in how doable they'd actually be in 14 weeks. Below is the comparison, covering the five things the assignment wants: feasibility, data/tool availability, technical risk, novelty, and how well it fits my background.

Candidate	Description	Feasibility (14 wks)	Data/Tool Availability	Technical Risk	Novelty	Fit with Background
A — Adversarial ML Attacks & Technical Countermeasures	Narrow, engineering-focused review of attacks on ML models (evasion, poisoning) and low-level defenses.	Medium. Heavily studied already, so hard to say something new in one term.	High. Plenty of literature, but very saturated.	Low technical risk, but real risk of just repeating existing surveys.	Low. This angle already has many existing technical surveys.	Good, but narrower than my actual interest in risk and governance.
B — Comparative Case Study of AI Governance in 2–3 Organizations	Empirical, qualitative comparison of real organizations' AI governance policies and practices.	Low. Needs organizational access, interviews, likely IRB review.	Low or uncertain. Depends on companies agreeing to take part.	High risk of stalling out if access falls through mid-term.	High if it works out, but timeline risk makes that unlikely.	Good fit, but the timeline risk is too high for a 14-week piece.
C — AI-Driven Cybersecurity Risk Management: Systematic Review + Framework (Selected)	Literature-based systematic review synthesizing threats, controls, and governance into one conceptual risk-management framework.	High. Fully literature-based, no dependency on outside access.	High. Confirmed by an early search: dozens of relevant recent reviews and frameworks	Low. Main risk is literature volume, manageable with a clear screening protocol.	Medium-high. 2026 reviews flag the ethics/compliance-vs-controls split as unresolved; closing that gap is the	Strong fit. Builds directly on my cybersecurity background and interest in governance and risk.

Candidate	Description	Feasibility (14 wks)	Data/Tool Availability	Technical Risk	Novelty	Fit with Background
			already located.		contribution here.	

Selection Rationale

I went with Candidate C. Honestly, it wasn't that close once I actually laid the three out side by side. It's the only one I can finish entirely through literature I already have access to through the university, which gets rid of the biggest risk with A and B: needing something outside my control to work out, whether that's finding something genuinely new in a field that's already crowded, or a company agreeing to hand over its data. It also has the clearest gap, and I'm not just saying that to justify my own pick. Recent reviews from 2025 and 2026 (Sources 2 and 3 in the Bibliography) already point out that AI governance research splits into two camps, ethics/compliance on one side, technical/threat on the other, and almost nothing connects them. Closing that gap is what this thesis is actually trying to do, and I think it's realistic in 14 weeks.

It also lines up with what CISC 699 wants: go from a broad interest to something scoped and doable. Nothing here depends on an organization letting me in halfway through the term, and I end up with one finished piece of scholarship instead of an open-ended build that could stall out. I'm hoping it also fits a supervisor whose background is more cybersecurity risk and governance than, say, applied machine learning engineering, but that's something I actually need confirmed at Discussion 1, not just assumed (see the Briefing Note).

As for timing, the six steps break into roughly two-week chunks that match the term's pace reasonably well, and none of them depend on someone else who could slow things down. That was really my worry with Candidate B.

3. Research Project Charter

Everything the charter needs is in this one section now, in the order the assignment lists it. The professor's feedback on the first round called out this stuff being scattered around, so I made a point of fixing that here.

Working Title

AI-Driven Cybersecurity Risk Management: A Systematic Review of Emerging Threats, Security Controls, and Governance Challenges.

Problem

See Section 1 for the full statement. In short: organizations are adopting AI faster than they can govern it, and the literature is heavy on describing AI threats but light on consolidated guidance connecting those threats to controls and governance.

Goals

- Complete a systematic review of the AI-cybersecurity risk literature using a documented, repeatable search and screening process.
- Identify and clearly state a research gap that existing reviews haven't already closed.
- Build a conceptual framework connecting AI risk identification to control selection to governance and compliance.
- Validate that framework against established, real-world frameworks instead of leaving it purely theoretical.
- Produce a final thesis document that meets the program's academic standards and is genuinely useful to practitioners, not just an academic exercise.

Intended Scholarly Artifact

Just to be clear up front: this is a theory and literature thesis, not a software build. What I'm actually producing is the thesis itself, a systematic literature review with a documented search and screening protocol, plus a conceptual risk-management framework (AI adoption to risk identification to risk assessment to control selection to governance and compliance to continuous monitoring), checked against established frameworks like the NIST AI Risk Management Framework, ISO/IEC 42001, and the NIST Cybersecurity Framework 2.0. I'm not writing code or training a model. The method is qualitative synthesis of what's already published, not programming.

On the data side, I'm only working with published academic and grey literature, pulled through university-licensed databases (Scopus, Web of Science, IEEE Xplore, ACM Digital Library) and standards or regulatory sites (NIST, ISO, ENISA, the EU). No APIs, no AI/ML models, no original software, no computational systems of any kind. The only real aids are a reference manager for keeping citations straight, and disclosed generative-AI help for brainstorming search terms and drafting, not for analysis or coding.

Likely Users / Stakeholders

- CISOs and IT risk managers who need a practical way to connect AI risks to concrete controls.
- Governance and compliance officers responsible for AI oversight and regulatory alignment.
- Boards and executives who sponsor AI adoption and are ultimately accountable for its risks.
- Standards and policy bodies who could use a synthesized picture of the literature.
- Future researchers who can build on, test, or extend the proposed framework.

Scope Boundaries

- Included: peer-reviewed literature and named frameworks, specifically the NIST AI RMF, ISO/IEC 42001, NIST CSF 2.0, and the EU AI Act, on AI-related cybersecurity risk, controls, and governance, mostly from 2019 to 2026.
- Included: organizational and enterprise-level AI use, not individual consumer AI risk.
- Excluded: primary data collection from real organizations. This is a literature-based review, not an empirical case study, and no primary human-subject data is collected.
- Excluded: low-level algorithmic security engineering, like designing a new adversarial defense algorithm. The focus stays at the risk-management and governance layer.
- Excluded: operationally actionable attack detail beyond what already exists in vetted public taxonomies like MITRE ATLAS. Threats are described taxonomically, not as how-to detail (see Section 6).

Assumptions

- Sufficient literature exists to support a rigorous review, already shown by preliminary searches returning dozens of relevant, recent (2023–2026) sources.
- University-accessible databases (library-licensed and open-access) will be enough; no paid datasets or proprietary tools are required.
- The supervisor will accept a systematic review plus conceptual framework as a valid deliverable, even though the course materials are written in build-an-artifact language.
- Disclosed generative-AI assistance for drafting and search-string brainstorming is permitted under the department's integrity policy, pending confirmation (see Section 6).

Constraints

- A fixed 14-week term, with the seven course milestone dates as hard stops.
- One student researcher, working without a data-collection or development team.
- English-language sources only, reachable through university-licensed or open-access channels.
- No budget for paid datasets, proprietary software, or compensated participants.

Risks

- The literature could get overwhelming fast. I'm dealing with that by setting clear inclusion/exclusion rules and category tags from week one, instead of just reading everything I come across.
- This field moves fast, NIST's Cyber AI Profile is still a draft as I write this. I'm picking a fixed search cutoff date partway through and noting anything major that comes out after that separately.
- It would be easy to drift into technical ML security detail that isn't really the point. I'm keeping every source and every piece of the framework tied to the risk-management and governance question, not the algorithm underneath it.
- Listing AI attack techniques in one place could actually help a malicious actor if I'm too specific. I'm sticking to the MITRE ATLAS taxonomy and describing things categorically, never operationally, and my advisor is reading the methodology chapter before it's finalized.
- The sources I'm finding lean English-language and vendor-sponsored, which could bias the conclusions. I'm searching multiple databases, deliberately pulling in non-U.S. sources like ENISA, and I'll be upfront about this limitation in the thesis itself.

Success Criteria

- A completed review following a documented protocol, covering an estimated 40 to 80 sources after screening.
- A clearly stated research gap, distinct from what existing reviews already say and backed by evidence.
- An internally consistent conceptual framework, compared against at least two established frameworks.
- A concluding discussion of the thesis's contribution and directions for future research.
- A final document meeting the program's formatting and citation requirements, approved by the supervisor at each milestone.

Supervisor Decisions Needed

Quick summary here; the full discussion is in Section 8, and the ethics-specific items also show up again in Section 6.

- Confirm a systematic review plus framework satisfies the deliverable requirement for a thesis-track student.
- Confirm source scope: peer-reviewed only, or peer-reviewed plus grey literature.
- Confirm citation style (APA 7 assumed).
- Confirm expected bibliography depth (roughly 40 to 80 screened sources).
- Confirm the validation bar: published frameworks, or organizational validation.
- Confirm whether formal IRB exemption sign-off is needed, and submit that request if so.
- Confirm the department's generative-AI disclosure policy.
- Confirm the final inclusion/exclusion criteria before full-scale screening begins.
- Confirm how much technical specificity is acceptable when describing AI attack techniques.

4. Feasibility Review

Feasibility for a project like this really comes down to two things: can I get to enough literature, and do I have enough time. There's no compute or software angle to worry about. Section 5 has the actual evidence for this, not just me saying it'll probably be fine.

Literature and Source Availability

I ran an initial search across the usual academic tools and publisher databases (Elsevier/ScienceDirect, Springer, MDPI, official NIST publications, among others) and turned up well over a dozen directly relevant reviews, frameworks, and recent papers within the first hour, several from as recently as 2025 and 2026. That tells me the topic is current and well covered, without being so saturated that there's nothing left for me to add.

Research Method (the equivalent of a "computational method" here)

Since there's no software pipeline, the closest thing I have to a method is the review process itself. I'm using a PRISMA-based approach (Page et al., 2021): pick search terms and databases ahead of time, screen everything against clear rules for what's in and what's out, tag each included source with one of the seven categories below, then pull it all together into the framework. I validate the framework by checking it against the NIST AI RMF, ISO/IEC 42001, and the CSF 2.0, not through any kind of technical test.

Time Constraints

Roughly, the six steps map onto a 14-week term like this: weeks 1 to 3 for search and organization, weeks 4 to 5 for pinning down the research gap, weeks 6 to 8 for sorting findings into the categories below, weeks 9 to 10 for building the framework, weeks 11 to 12 for validating it, and weeks 13 to 14 for the conclusion and finishing the draft. This is a working plan, not a fixed schedule, and it'll get adjusted after Discussion 1.

Category Structure for Analysis

Findings will be organized into the following categories, which also double as the coding scheme for the bibliography and the screening tracker in Section 5:

- AI-related threats
- Privacy risks
- Security risks
- Human/organizational risks
- Governance risks
- Compliance risks
- Security controls

5. Technical Readiness Evidence

No software here, so instead of a repo and code, this is the research equivalent: a project overview, a working folder structure, a reference tracker, a research log, a scope diagram, and a source dependency table — all already set up, not just planned.

Evidence Already in Place

- Project overview: a one-page README kept at the top of the workspace, stating the project type (thesis, no software component), current status (launch phase, initial screening underway), and contents (literature by category, screening tracker, framework drafts, chapter drafts).
- Folder structure: a working directory already split into Literature (seven risk-category subfolders), Screening Tracker, Framework Drafts, Meeting Notes, Chapter Drafts, and an Ethics folder holding the stakeholder map and risk register.
- Reference manager: sources stored and tagged in a Zotero library, each tagged by category and logged with an include/exclude decision and reason — for example, Albulayhi & Alkhalifah (2026), tagged "Governance risks" and "Security controls," included because it supports the research gap directly.
- Research log: a running journal, started the week of September 8, 2026 — course milestones logged, initial searches run to check feasibility, the Zotero library and folders set up, and the problem statement, candidate directions, and Discussion 1 draft all completed.

Scope Diagram (Conceptual Framework)

This is the conceptual framework the thesis will build and validate, and also its scope boundary: everything in the flow is in scope; lower-level detail, like how a specific ML defense algorithm works, is out of scope.

AI adoption → Identify cybersecurity risks → Risk assessment → Select security controls → Governance & compliance → Continuous monitoring

Source Dependency Table

The closest equivalent to an API table: the databases and tools the review depends on, and access status.

Database / Source	Access Status	Role
IEEE Xplore, ACM Digital Library, ScienceDirect	Access via university library	Core bibliographic databases for technical and security papers.
Scopus / Web of Science	Access via university library	Citation tracking and search coverage checks.
Google Scholar	Freely accessible	Supplementary and grey-literature searching.
NIST and ENISA publications	Freely accessible	Governance frameworks used for validation (AI RMF, CSF 2.0); ENISA adds a non-U.S. source.
Zotero	Installed and configured	Reference manager: storage, tagging, APA export.

6. Ethics and Integrity Readiness

This is a condensed version of my separate Discussion 1 post (Ethics Readiness and Evidence of Preparation). I'm including it here too so this packet doesn't depend on someone finding that other post to make sense.

Ethics Screening

I ran this project through eight standard ethics categories, and six of them actually applied. Data ownership and permitted use, since the corpus is built on licensed databases and copyrighted material. Bias and representational harm, because English-language and vendor-sponsored sources are probably overrepresented. Transparency and reproducibility, since PRISMA basically requires a search trail anyone could follow. Dual-use risk, because listing AI attack techniques in one place could actually help someone if I get too specific. IP and licensing, since quoting and attribution rules apply to every source. And academic integrity around AI use, because any AI-assisted drafting has to be disclosed and checked. Privacy wasn't really relevant here; I'm not collecting personal data, the only thing to watch for is if the literature itself names identifiable breach victims.

Three Priority Risks

- Dual-use risk: if I describe AI attack techniques too specifically, I could basically hand attackers a cheat sheet. I'm handling this by sticking to the categories in the MITRE ATLAS taxonomy instead of writing anything operational, and having my advisor read the methodology chapter before it goes anywhere.
- Bias: leaning too hard on English-language or vendor-funded sources could skew what I end up concluding. I'm pulling from multiple databases, deliberately including non-U.S. sources like ENISA, and I'll just say outright in the thesis where this limitation still exists.
- AI disclosure: not disclosing or double-checking AI-suggested text and citations is an integrity problem waiting to happen. I'm keeping a running log (Section 9), checking every citation against the actual source, and following whatever the program's AI policy turns out to be.

Readiness Declaration

Here's where things actually stand. Already confirmed: the topic and scope, that there's no primary human-subject data involved, that PRISMA 2020 is the methodology, and which governance frameworks I'm reviewing (NIST AI RMF, ISO/IEC 42001, NIST CSF 2.0, the EU AI Act).

Still just assumptions right now: that I won't need full ethics-board review, that disclosed AI help is fine, that my database access holds up for the whole review period, and that mixing academic and grey literature is okay with my committee.

Genuinely unresolved, and here's my plan for each one: whether I even need a formal IRB exemption (filing that request two weeks before this is due), what the department's actual policy is on disclosing AI use (checking with the graduate integrity office, same deadline), what my final inclusion and exclusion criteria will be (advisor meeting a week out), and how specific I'm allowed to get when describing attacks (advisor and ethics reviewer reading the draft methodology chapter).

And before this is actually due, I still need to file the IRB exemption request, lock down the PRISMA search protocol, turn the stakeholder map and risk register into a real appendix, and get the AI disclosure requirement confirmed in writing instead of just assumed.

Decision Question

Here's honestly where I'd pull back or rethink things. If answering my research questions meant getting into attack detail more specific than what's already public in something like MITRE ATLAS, detail that could actually help a bad actor, I'd scale back to defensive controls and governance only, no threat cataloguing. Same story if an ethics board came back requiring full review and turned up a privacy issue in the secondary case material I couldn't work around, or if I ended up needing data outside what I actually have licensed access to.

7. Annotated Bibliography

There are fourteen sources here now. It was seven before I wrote Discussion 1, but that process turned up several governance and ethics sources I couldn't really leave out. Between all of them, they cover the general AI-in-cybersecurity literature, AI governance

frameworks, IT risk governance, one binding regulation plus a few voluntary frameworks I'll validate against, a real-world case that motivates the whole thing, a methods source, a threat taxonomy, and the academic-integrity guidance behind how I'm disclosing AI help in this packet. All APA 7th edition. I haven't read every source cover to cover yet, these annotations come from abstracts and public summaries, so that's still on my list before I actually lean on them in the literature review.

Kaur, R., Gabrijelčič, D., & Klobučar, T. (2023). Artificial intelligence for cybersecurity: Literature review and future research directions. *Information Fusion, 97*, Article 101804. https://doi.org/10.1016/j.inffus.2023.101804

A broad map of how AI shows up across cybersecurity in general. Useful as an anchor for the "AI-related threats" category, and it makes my point for me: most existing work treats AI-in-security broadly rather than through a structured risk-management lens.

Albulayhi, O., & Alkhalifah, A. (2026). AI governance risk tiering for sustainable digital infrastructure: A systematic review of cybersecurity frameworks. *Sustainability*, *18*(6), Article 2986. https://doi.org/10.3390/su18062986

Probably the single most important source for the research gap: a very recent review (95 studies from 450 records) finding AI governance frameworks split into ethics/privacy-focused and compliance/risk-focused camps, with almost nothing connecting either to operational controls.

Mokgathi, M., & Ogundele, O. S. (2026). Information technology (IT) risk governance and cybersecurity threats: A systematic literature review. In *Embracing technological agility in accounting and business* (Vol. 2). Springer.

A PRISMA-based review of IT risk governance strategies (2016–2025) calling for governance models to adapt to newer technologies including AI. Its method is a good template for my own search and screening.

National Institute of Standards and Technology. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100-1). U.S. Department of Commerce. https://doi.org/10.6028/NIST.AI.100-1

The most widely cited official U.S. framework for managing AI risk in general. One of the benchmarks I'll validate my own framework against.

National Institute of Standards and Technology. (2025). *Cybersecurity framework profile for artificial intelligence (Cyber AI profile)* (NIST IR 8596, Initial Preliminary Draft). U.S. Department of Commerce.

A late-2025 draft mapping AI-specific considerations onto the NIST CSF 2.0's six functions. Useful for the "security controls" category and as a second validation benchmark.

Zeijlemaker, S., Lemiesa, Y. K., Schröer, S. L., Abhishta, A., & Siegel, M. (2025). How does AI transform cyber risk management? *Systems*, *13*(10), Article 835. https://doi.org/10.3390/systems13100835

Combines a literature review with a real case study, the Colonial Pipeline attack, showing how AI shifts cyber risk toward compounding, adaptive patterns. Gives me a concrete example for the introduction.

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., … Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ, 372*, Article n71. https://doi.org/10.1136/bmj.n71

The standard reporting guideline for systematic reviews, and the basis for my own search, screening, and inclusion/exclusion process.

International Organization for Standardization. (2023). *Information technology — Artificial intelligence — Management system* (ISO/IEC 42001:2023). https://www.iso.org/standard/42001.html

The first certifiable AI management system standard, picked up while drafting Discussion 1. Anchors part of the governance-layer analysis and is now a validation benchmark too.

National Institute of Standards and Technology. (2024). *The NIST Cybersecurity Framework (CSF) 2.0* (NIST CSWP 29). U.S. Department of Commerce. https://doi.org/10.6028/NIST.CSWP.29

Its Govern function is a close match for the governance layer this thesis builds toward; the third validation benchmark alongside the AI RMF and ISO/IEC 42001.

European Parliament and Council of the European Union. (2024). *Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)*. Official Journal of the European Union. https://eur-lex.europa.eu/eli/reg/2024/1689/oj

The one binding hard-law source in this list, as opposed to a voluntary framework. Matters for the "compliance risks" category specifically.

MITRE Corporation. (n.d.). *MITRE ATLAS: Adversarial threat landscape for artificial-intelligence systems*. Retrieved September 2026, from https://atlas.mitre.org

A publicly vetted taxonomy of AI attack techniques at a categorical, not operational, level. This is what lets the thesis discuss threats seriously without the dual-use risk flagged in Section 6.

Committee on Publication Ethics. (2024). *Authorship and AI tools* [Position statement]. https://publicationethics.org/guidance/cope-position/authorship-and-ai-tools

Note: my Discussion 1 draft cited this as 2023, but it was published August 20, 2024, so I corrected the year here. This is the actual standard behind how I disclose AI-assisted drafting in the AI Usage Log.

International Committee of Medical Journal Editors. (n.d.). *Defining the role of authors and contributors*. Retrieved September 2026, from https://www.icmje.org/recommendations/browse/roles-and-responsibilities/defining-the-role-of-authors-and-contributors.html

Specific to medical journals, but the underlying principle, that the named human author stays accountable regardless of what assistance was used, is the standard I'm holding this packet to.

Association for Computing Machinery. (2018). *ACM code of ethics and professional conduct*. https://www.acm.org/code-of-ethics

The general professional-ethics backdrop for this project: honest reporting, avoiding harm. Most directly relevant to the dual-use risk around describing attack techniques.

8. Supervisor Briefing Note — Questions for Discussion 1

These are the things I actually need an answer on before the next milestone (some of this repeats the Charter and the ethics section, just gathered here for the actual conversation):

1. Track fit: can you confirm a systematic literature review plus a conceptual framework counts as this course's required deliverable for a thesis-track student?

2. Source scope: should the review include grey literature alongside peer-reviewed sources, or stay peer-reviewed only?

3. Citation style: confirming APA 7th edition is correct (used as the default here).

4. Expected depth: is 40 to 80 screened sources reasonable, or does the program expect a different scale?

5. Validation bar: is comparison against published frameworks enough, or is organizational validation expected?

6. Supervisor fit: can you confirm your background lines up with AI governance/cybersecurity risk?

7. IRB and disclosure: does a literature-only review need formal IRB exemption, and what's the department's generative-AI disclosure policy?

8. Technical specificity: how much detail is acceptable when describing AI attack techniques?

9. AI Usage Log

Here's the actual, detailed record of how I used AI (Claude) while putting this packet together. I'm keeping this specific on purpose, since the professor flagged that a vague "I used AI" line isn't good enough, and it lines up with the COPE and ICMJE guidance cited in Section 7. Every entry covers the tool, when it happened, why, what I asked for, what I actually used, what it touched, how I checked it, and what I changed myself afterward.

Entry 1 — Initial Drafting (Week of September 8, 2026)

AI Tool: Claude (Anthropic AI assistant)

Purpose: Structure and draft the launch packet, and find candidate literature sources.

Prompt / Task Summary: Gave Claude the thesis topic and the assignment's tasks and rubric; asked it to draft a problem statement, compare three candidates, build a charter, feasibility review, workspace plan, and find at least five credible sources.

Output Used: Initial drafts of the problem statement, candidate comparison, charter, feasibility review, workspace section, and six sources with draft annotations.

Affected Section / Artifact: Sections 1 through 8, first full draft.

Human Verification: Confirmed the topic and final candidate selection myself; had Claude pull and confirm citation details for each source rather than accept a generated citation; still need to read each source in full.

Final Student Revisions: Edited the language throughout for accuracy and voice.

Entry 2 — Revision After Class-Wide Feedback (September 9, 2026)

AI Tool: Claude (Anthropic AI assistant)

Purpose: Revise the packet in response to the professor's class-wide feedback on the first, ungraded round.

Prompt / Task Summary: Gave Claude the feedback email describing common deficiencies (incomplete comparisons, scattered charter elements, missing technical-readiness evidence, weak bibliographies, generic AI logs) and asked for each point to be addressed.

Output Used: Added a Novelty column and course-outcome/supervisor-fit/timeline language; reorganized the Charter; added Technical Readiness Evidence; added a PRISMA source; rebuilt this Log into field-by-field entries; removed all colour and page numbers.

Affected Section / Artifact: Candidate comparison, Charter, Technical Readiness Evidence, Bibliography, this Log.

Human Verification: Compared each change against the feedback email line by line; independently checked the added PRISMA citation.

Final Student Revisions: Reviewed the revised language for tone and accuracy.

Entry 3 — Rubric Re-check and Language Pass (September 9, 2026)

AI Tool: Claude (Anthropic AI assistant)

Purpose: Re-check the packet against the rubric line by line and rewrite the prose to sound like my own voice.
Prompt / Task Summary: Asked Claude to confirm the top rubric level is met for each criterion, and to reduce repeated sentence patterns and em-dash overuse.
Output Used: Added a "Goals" list to the Charter and a "Research Method" subsection to Feasibility; rewrote most connecting prose, including all bibliography annotations at the time.
Affected Section / Artifact: Charter, Feasibility Review, general language throughout.
Human Verification: Checked the rewrites against the actual rubric text, criterion by criterion.
Final Student Revisions: Confirmed the changes matched my own understanding of the rubric.

Entry 4 — Integrating Discussion 1 (September 9, 2026)

AI Tool: Claude (Anthropic AI assistant)
Purpose: Fold the governance frameworks and ethics analysis from my Discussion 1 draft into this packet.
Prompt / Task Summary: Gave Claude my full Discussion 1 draft and asked what Assignment 1 needed to stay consistent with it.
Output Used: New frameworks and a Data/APIs statement in the Charter; two new risks; four new Supervisor Decisions items; a new Ethics and Integrity Readiness section; seven new bibliography entries; an updated dependency table.
Affected Section / Artifact: Problem Statement, Charter, Technical Readiness Evidence, Ethics section, Bibliography, Briefing Note, this Log.
Human Verification: Had Claude verify each new citation independently, which caught a wrong year on the COPE citation in my own Discussion 1 draft (2023 vs. the actual 2024).
Final Student Revisions: Will confirm the corrected COPE year and this whole update against my own Discussion 1 submission before treating either as final.

Entry 5 — Length Reduction (September 9, 2026)

AI Tool: Claude (Anthropic AI assistant)
Purpose: Shrink the packet to about nine pages for submission, without dropping any required content.
Prompt / Task Summary: Asked Claude to reduce the page count through formatting (smaller font and margins, tighter spacing) and trimming redundant wording, not by deleting required sections.
Output Used: Tighter margins, spacing, and font size throughout; shortened some sentences and bullets that repeated information already stated elsewhere.
Affected Section / Artifact: Formatting throughout; light text trims in several sections.
Human Verification: Compared this version against the previous one section by section to make sure no required rubric element was cut, only shortened.
Final Student Revisions: Will do a final read to make sure nothing feels rushed or unclear after the trim.

Entry 6 — Voice and Tone Pass (September 9, 2026)

AI Tool: Claude (Anthropic AI assistant)
Purpose: Rewrite the packet's prose so it reads like my own voice instead of a polished, generic AI-drafting style.
Prompt / Task Summary: Told Claude the writing still felt too AI-generated and asked for a pass that sounds more like an actual student wrote it: fewer perfectly balanced sentences, more contractions and asides, less template-like phrasing (especially the "Risk: … Mitigation: …" pattern and the Confirmed/Assuming/Unresolved lists).
Output Used: Rewrote most of the connecting prose across the packet in a looser, more conversational voice, and loosened some of the labeled bullet structures into fuller sentences while keeping the same information.
Affected Section / Artifact: Problem Statement, candidate rationale, Charter intros, Feasibility, Technical Readiness intro, the Ethics section, and the Bibliography and Briefing Note intros.
Human Verification: Read through the rewritten version myself to make sure the looser phrasing didn't accidentally change what any of it actually claims.
Final Student Revisions: Will read it once more out loud before submitting, since that's usually how I catch the parts that still don't sound like me.

Entry 7 — Condensing Technical Readiness Evidence (September 10, 2026)

AI Tool: Claude (Anthropic AI assistant)
Purpose: Shorten the Technical Readiness Evidence section without dropping any of the required evidence types.
Prompt / Task Summary: Asked Claude to make the Technical Readiness Evidence section shorter and more summarized.
Output Used: Combined the README stub, folder-structure listing, reference-manager note, and research-log entry into one compact evidence list; trimmed the scope diagram to a single line; reduced the source dependency table from nine rows to six by grouping related databases.
Affected Section / Artifact: Section 5, Technical Readiness Evidence, and this Log.
Human Verification: Checked that every evidence type the rubric asks for (project overview, folder structure, log, diagram, dependency table) is still present, just written more briefly.
Final Student Revisions: Will confirm with my supervisor that the shorter version still reads as concrete evidence rather than a plan.

Entry 8 — Fixed a Section Numbering Gap (September 13, 2026)

AI Tool: Claude (Anthropic AI assistant)

Purpose: Final completeness check before treating this packet as done.

Prompt / Task Summary: Asked Claude to confirm the assignment was complete.

Output Used: Caught that an earlier revision left a numbering gap: the document jumped from Section 7 straight to a section labeled 9, with no Section 8. Renumbered the Supervisor Briefing Note from 9 to 8 and the AI Usage Log from 10 to 9, and fixed the two in-text cross-references that pointed to the old numbers.

Affected Section / Artifact: Section headings 8 and 9, the cross-reference in Section 3, and the cross-reference in Section 6.

Human Verification: Read through the section list top to bottom to confirm the numbers now run 1 through 9 with nothing skipped, and checked both fixed cross-references point to the right section.

Final Student Revisions: None needed beyond this fix; I'll still give the whole document one more read before submitting.

Reflection: AI assistance was used for structure, drafting speed, source discovery, formatting, and tone editing, not for the actual research or ethics judgment calls. Those decisions are mine, and I take responsibility for verifying every source and claim in this packet before submission.